

INFORMATION TECHNOLOGY INSTITUTE

LAB 9

Two-Stage Miller OTA





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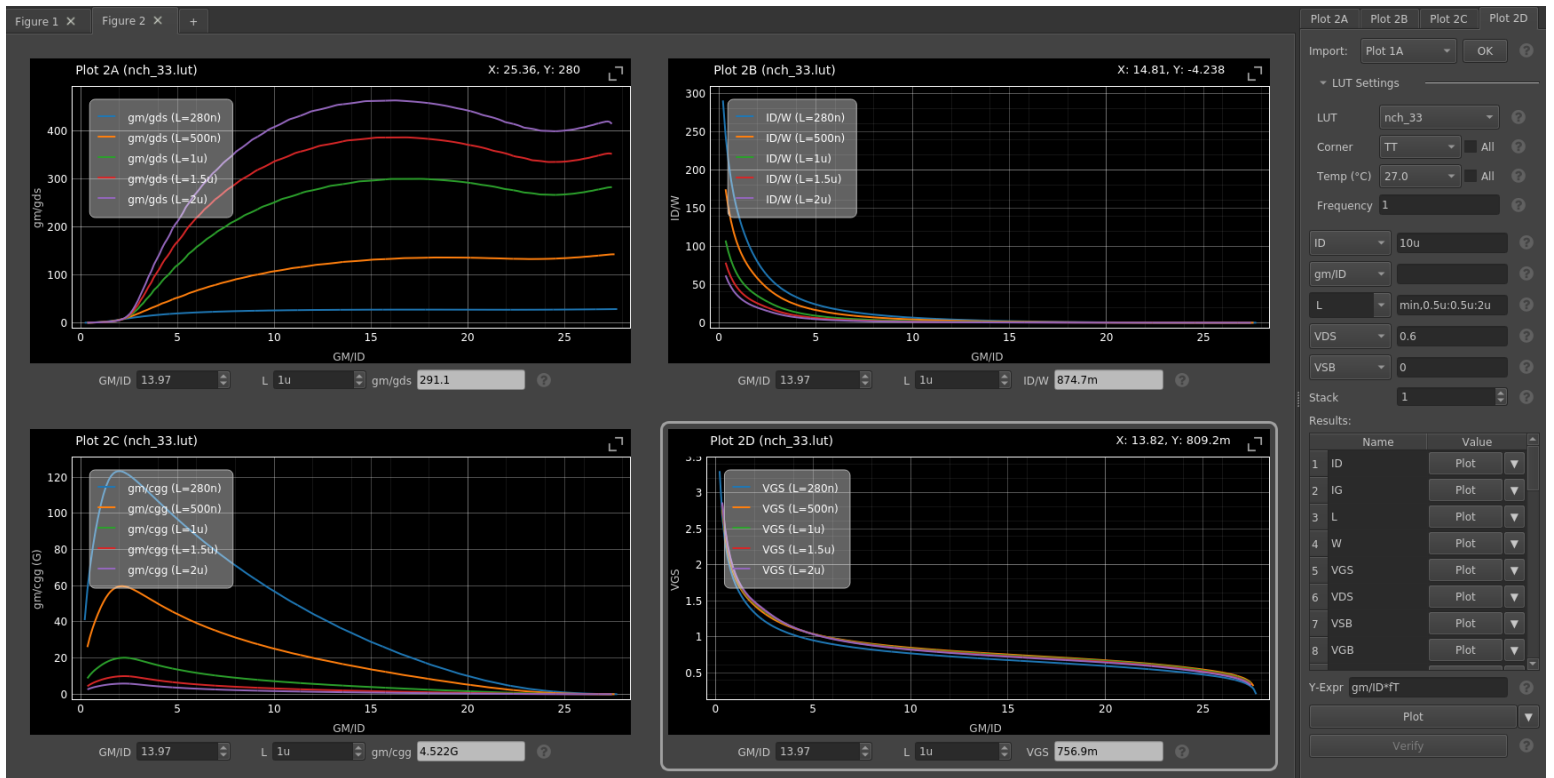


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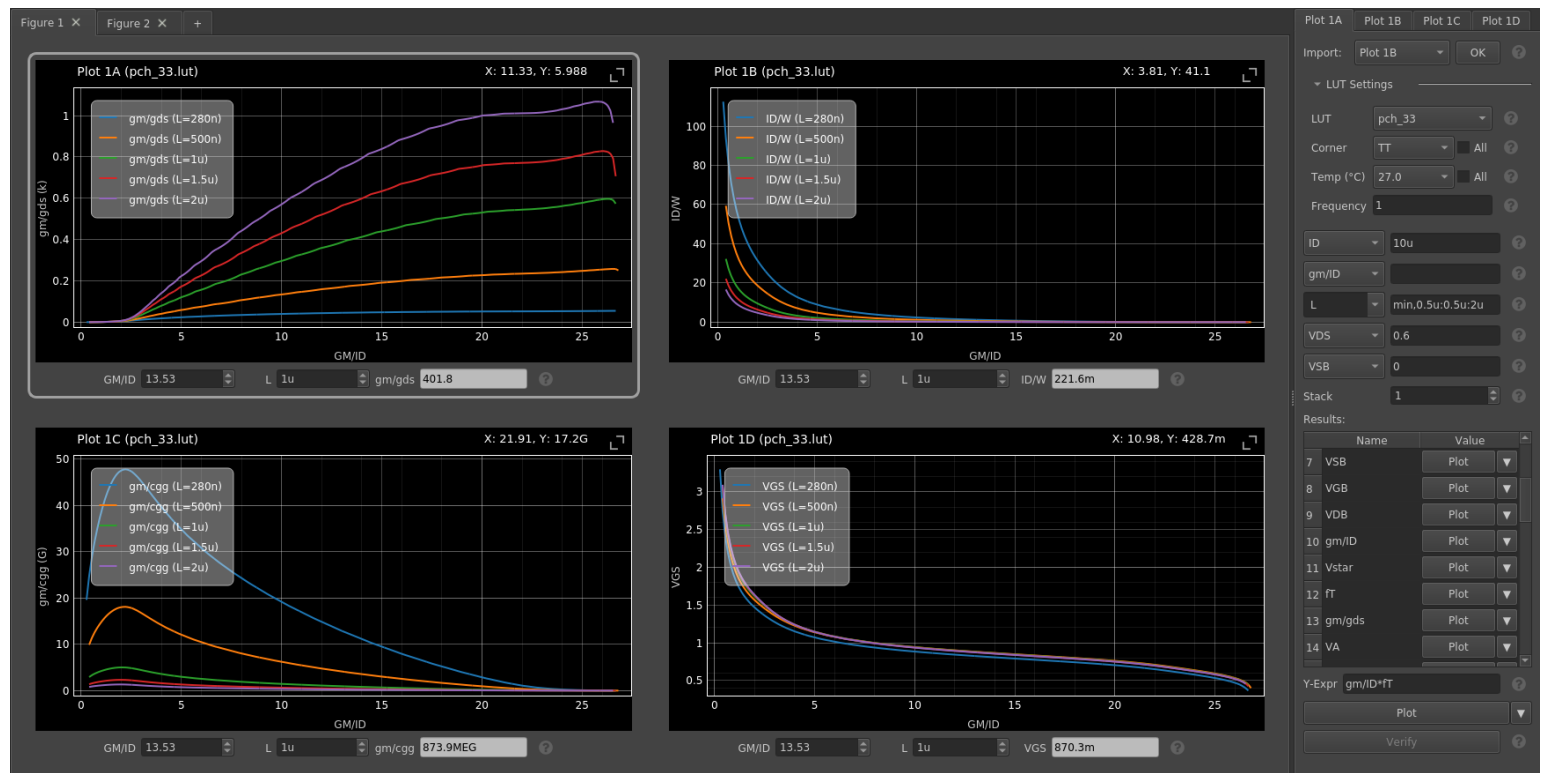


PART 1: gm/ID Design Charts

Nmos



Pmos





PART 2: OTA Design

Technology	0.13um	0.18um
Supply voltage	1.2V	1.8V
Static gain error	$\leq 0.05\%$	$\leq 0.05\%$
CMRR @ DC	$\geq 74\text{dB}$	$\geq 74\text{dB}$
OTA current consumption	$\leq 60\mu\text{A}$	$\leq 60\mu\text{A}$
CMIR – high	$\geq 0.6\text{V}$	$\geq 1\text{V}$
CMIR – low	$\leq 0.2\text{V}$	$\leq 0.2\text{V}$
Output swing	0.2 – 1V	0.2 – 1.6V
Load	5pF	5pF
Buffer closed loop rise time (10% to 90%)	$\leq 70\text{ns}$	$\leq 70\text{ns}$
Slew rate (SR)	$5\text{V}/\mu\text{s}$	$5\text{V}/\mu\text{s}$

First we will assume the CC I first assumed it by 2.5p f but then changed it to 2p f to meet the slew rate spec at the current I choose.

Then we will try to find the gm of the input pair by using the buffer CL risetime as the when connected as a buffer we can assume that the time constant of the system is the UGF of the Aol and approximate it a first order system $UGF = gm/CC$ so the $\tau = CC/gm = 31.818\text{ns} = 2e-12 / gm$, so gm1,2 should be equal or less than $6.285e-5$ so it we can assume a $gm/id = 15$ for input pair at current equal 5uA but what wont this causes a larger gm than needed yes and this is actually good as the rise time will decreases

Then by the PM info and the gm1,2 we can find gm of the nmos second stage how we know that to achieve a PM greater than 70 we need $wp2 = 3 * Wu$ so to have a safe margin and have a better response for the system we will choose a ratio of 4 to be critically damped which will improve the PM and transient response $wu = gm1/CC$ while $wp2 = gm2/CL$ so we will need a gm that is ten times more the gm of the first stage this is actually great as we choose the first stage current in each branch = 5uA then the output we will have a current of 50uA so we can bias on the same gm/id reserving area



Now using the gain we will assume or target a first stage gain of 70 and second of 35 so to leave margin and as we want the first stage gain bigger as it helps in CMRR, we could have a design path in which we will assume that the load mosfet has the same VA as the input and targeted G_{mro} double what will the actual gain be but I choose another path to find the L or r_o we need to the second stage nmos should match the load nmoses or at least has a the same VGS so there won't be any internal systematic error which cause the output to hit the rail I preferred matching the nmos to load instead of just matching VGS because good designs depend on ratios and not nominal values and thus it will be more stable across corners as we said the second stage will have a g_m/id of 15 like input and also the load will have the g_m/id so and must have a lower r_o than the input because now the load mosfet is tied to the second stage nmos which will target it having a low r_o so it will have a low overall gain lower than the first stage.

By now we can use another spec to find another unknown the spec is the CMRR as specified the CMRR should be more than 5011V/V and we have assumed a first stage gain of 70 V/V so $AV_{cm} = 1/2 * g_{m3,4} * R_{SS} = 0.01467$ and the $g_{m3,4}$ will $g =$ have the same g_m of the input pair as they will have the same g_m/id which is around 73 to 75 μS so we will estimate an R_{SS} of 480k to achieve needed CMRR we know that the current mirroring device should have a lower g_m/id but here this not achievable as the CMIR high is 0.8v which leaves only one volt for the VGS of the input and the V^* of the CM device so I will choose $g_m/id = 10$ and I cant go any lower to increase the R_{SS}

After knowing the exact R_{SS} we can now find the exact L of the second stage Nmos which will help us achieve the gain distribution that we want in the second stage branch the current 50 μA while in the first stage CM it is 10 μA so the R_{SS} will be divided by 5 for the second stage which is equal to 96k and we said that the g_m of this stage will be around 750 μS so we the r_o for this Nmos is around 100k as the gain is $G_m(r_{op} || r_{on})$.

Now we can continue sizing the input pair we know that current is 10 times less than the second stage thus the load mosfets have $10 * r_o$ but the G_M of the input will be less by 10 times so to solve this problem we will say that the load mosfets have around 1Mohm r_o and the input then will need around 4M to 7M ohms to achieve the gain distribution needed and this I actually a bit weird because in



normal cases the r_o of the load is larger but here because it was tied to a transconducting mosfet it went the other way around .

We can use this rough constraints to get an initial sizing that we will later tune

CM device sizings

Plot 3A Plot 3B Plot 3C Plot 3D

Import: Plot 1A OK ?

▼ LUT Settings

LUT pmos_03v3 ?

Corner TT All ?

Temp (°C) 27.0 All ?

Frequency 1 ?

ID 10u ?

gm/ID 14 ?

r_o 480k ?

VDS 0.3 ?

VSB 0 ?

Stack 1 ?

Results:

	Name	TT-27.0
1	ID	10u
2	IG	N/A
3	L	380n
4	W	15.32u
5	VDB	300m
6	VGB	852.4m
7	VGS	852.4m
8	VDS	300m

Y-Expr gm/ID*fT ?

Plot ▼

Verify ?

Plot 5A Plot 5B Plot 5C Plot 5D

Import: Plot 1A OK ?

▼ LUT Settings

LUT pmos_03v3 ?

Corner TT All ?

Temp (°C) 27.0 All ?

Frequency 1 ?

ID 50u ?

gm/ID 14 ?

L 380n ?

VDS 0.3 ?

VSB 0 ?

Stack 1 ?

Results:

	Name	TT-27.0
1	ID	50u
2	IG	N/A
3	L	380n
4	W	78.6u
5	VDB	300m
6	VGB	850.2m
7	VGS	850.2m
8	VDS	300m

Y-Expr gm/ID*fT ?

Plot ▼

Verify ?



NMOS loads and second stage

Plot 2A Plot 2B Plot 2C Plot 2D

Import: Plot 1A OK ?

▼ LUT Settings

LUT: nmos_03v3 ?

Corner: TT All ?

Temp (°C): 27.0 All ?

Frequency: 1 ?

ID: 5u ?

gm/ID: 15 ?

L: 400n ?

VDS: vgs ?

VSB: 0 ?

Stack: 1 ?

Results:

	Name	TT-27.0
1	ID	5u
2	IG	N/A
3	L	400n
4	W	3.08u
5	VDB	733m
6	VGB	733m
7	VGS	733m
8	VDS	733m

Y-Expr: gm/ID*fT ?

Plot ▼

Verify ?

Plot 4A Plot 4B Plot 4C Plot 4D

Import: Plot 1A OK ?

▼ LUT Settings

LUT: nmos_03v3 ?

Corner: TT All ?

Temp (°C): 27.0 All ?

Frequency: 1 ?

ID: 50u ?

gm/ID: 15 ?

gm/gds: 90 ?

VDS: 0.9 ?

VSB: 0 ?

Stack: 1 ?

Results:

	Name	TT-27.0
1	ID	50u
2	IG	N/A
3	L	400n
4	W	30.64u
5	VDB	900m
6	VGB	731.3m
7	VGS	731.3m
8	VDS	900m

Y-Expr: gm/ID*fT ?

Plot ▼

Verify ?



Input pair sizing

What was actually fine tuned

I tuned the r_o of the nmos second stage so I can find the balance between the gain of the first stage and second stage as we know if we decrease the r_o of the second stage the load r_o will also decrease so I tuned it till I reach a point in which the r_o of the second stage is small but the load r_o is still acceptable and thus I won't need to increase the L of the input that much.

I also decreased the CC capacitor so it the Patristics keep the capacitance at this node Equals as if $cc=2p$ cc new value is 1.8pf And also we add a resistance R_z to get rid The feedforward zero which is equal to $1/GM2 = 1.353k$.

Plot 1A Plot 1B Plot 1C Plot 1D

Import: Plot 1A OK ?

LUT Settings

LUT pmos_03v3 ?

Corner TT All ?

Temp (°C) 27.0 All ?

Frequency 1 ?

ID 5u ?

gm/ID 15 ?

L 690n ?

VDS 0.6 ?

VSB 0 ?

Stack 1 ?

Results:

	Name	TT-27.0
1	ID	5u
2	IG	N/A
3	L	690n
4	W	20.4u
5	VDB	600m
6	VGB	850m
7	VGS	850m
8	VDS	600m

Y-Expr gm/ID*fT ?

Plot ▼

Verify ?

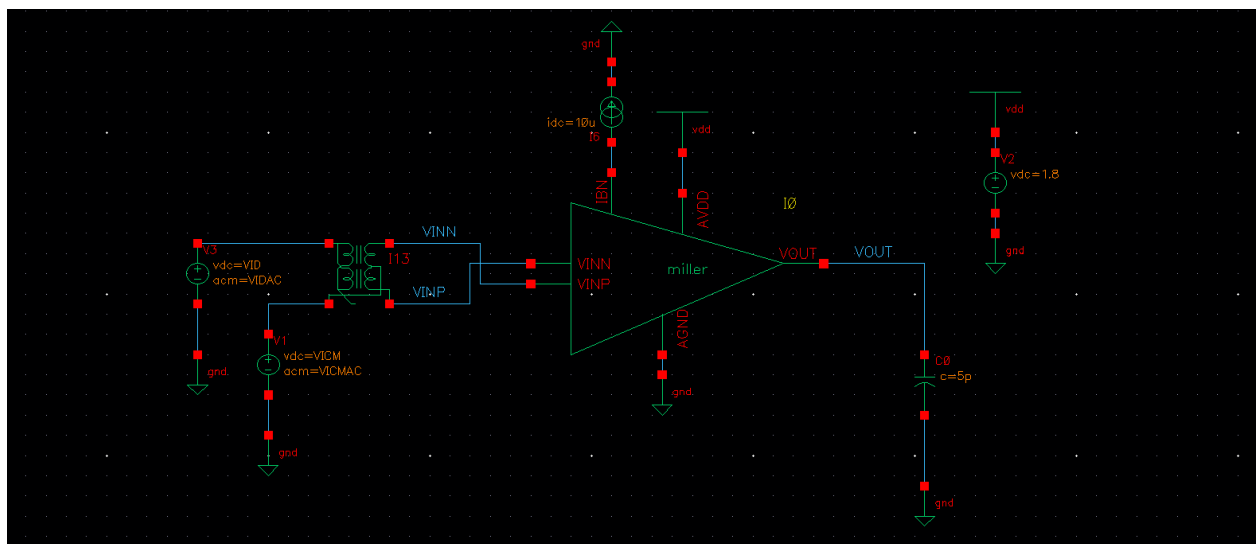
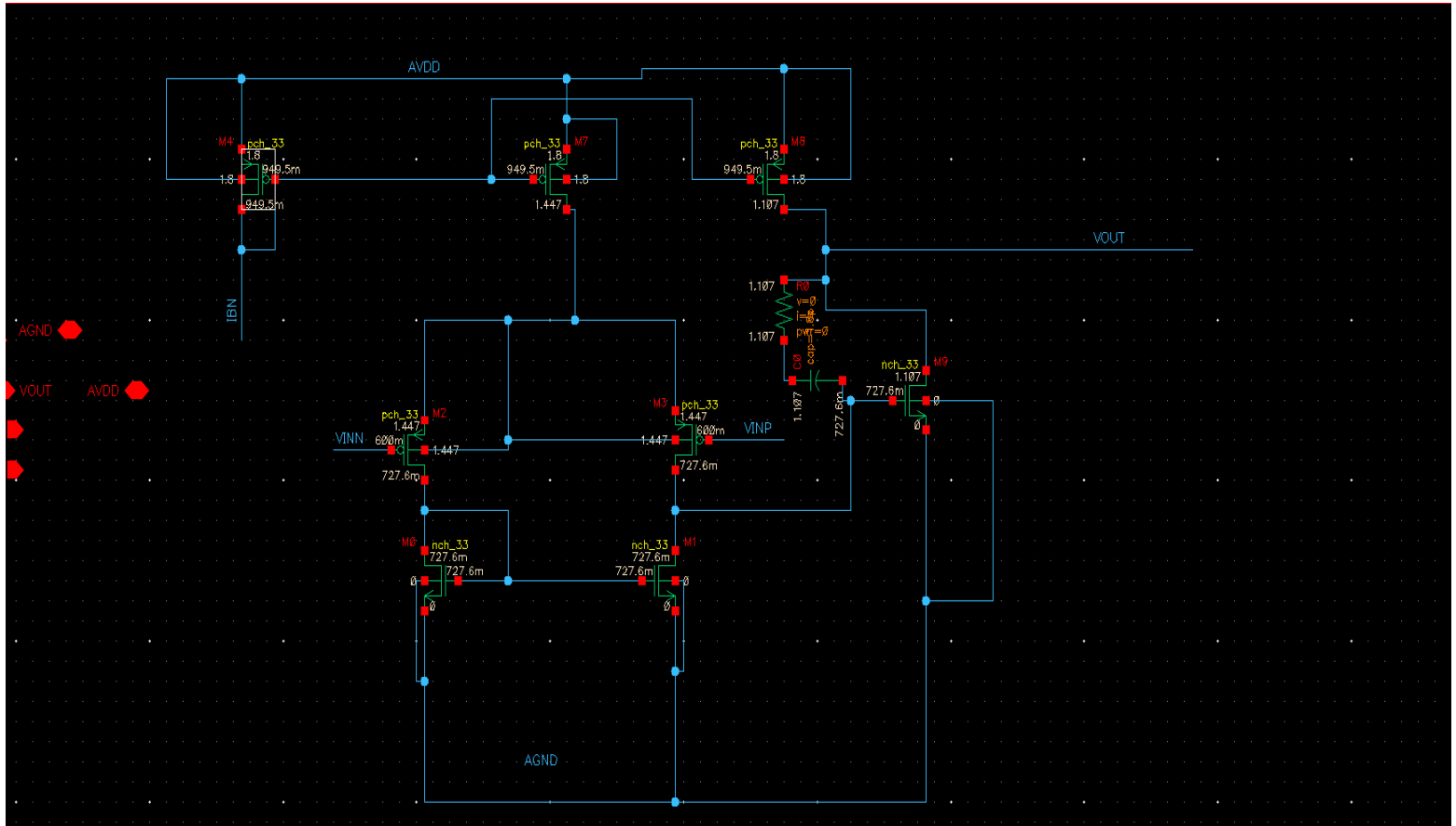
Transistor(s)	Function	Type	W	L	I _b	g _m /I _b	g _m	V _{gs}	V _{th}	V _{ov}	V _{sat}	V*
M0, M1	First-stage active load	NMOS	3.08 μm	400 nm	5 μA	14.76 V^{-1}	73.8 μS	733.0 mV	693.9 mV	39.1 mV	107.2 mV	135.5 mV
M2, M3	Differential input pair	PMOS	20.4 μm	690 nm	5 μA	15.00 V^{-1}	75.0 μS	850.0 mV	787.7 mV	62.3 mV	100.6 mV	133.3 mV
M4	Diode-connected 10- μA bias	PMOS	15.32 μm	380 nm	10 μA	13.82 V^{-1}	138.2 μS	852.4 mV	773.7 mV	78.7 mV	122.1 mV	144.7 mV
M7	10- μA first-stage current source	PMOS	15.32 μm	380 nm	10 μA	13.82 V^{-1}	138.2 μS	852.4 mV	773.7 mV	78.7 mV	122.1 mV	144.7 mV
M8	50- μA second-stage current source	PMOS	78.6 μm	380 nm	50 μA	13.99 V^{-1}	699.5 μS	850.2 mV	773.7 mV	76.5 mV	120.6 mV	142.9 mV
M9	Second-stage common-source device	NMOS	30.64 μm	400 nm	50 μA	14.78 V^{-1}	739.0 μS	731.3 mV	693.2 mV	38.1 mV	106.6 mV	135.3 mV





PART 3: Open-Loop OTA Simulation

Schematic of the OTA and bias circuit with DC node voltages clearly annotated.





- Yes the current in the input pair are exactly the same .
- DC level of the first stage is the VGS of the diode transistor due the output node of the first stage following the the diode connection node in common mode.
- Theoretically it should have been 0.9 but because it's a very high impedance node its voltage has changed as it is an ill-defined node and this is why we did the matching between the Nmos devices in this circuit however this can still be fixed by fine tuning the W of the nmos output stage but this may change the phase margin and change the gain distribution

Operating points

Gm

VIDAC	0.000
I0.M0.m0:gm (S)	71.27E-6
I0.M1.m0:gm (S)	71.27E-6
I0.M2.m0:gm (S)	71.82E-6
I0.M3.m0:gm (S)	71.82E-6
I0.M4.m0:gm (S)	141.7E-6
I0.M7.m0:gm (S)	133.2E-6
I0.M8.m0:gm (S)	714.5E-6
I0.M9.m0:gm (S)	747.5E-6

id

VIDAC	0.000
I0.M0.m0:id (A)	4.662E-6
I0.M1.m0:id (A)	4.662E-6
I0.M2.m0:id (A)	4.662E-6
I0.M3.m0:id (A)	-4.662E-6
I0.M4.m0:id (A)	-4.662E-6
I0.M7.m0:id (A)	-10.00E-6
I0.M8.m0:id (A)	-9.324E-6
I0.M9.m0:id (A)	-50.26E-6

Region

VIDAC	0.000
I0.M0.m0:region	2
I0.M1.m0:region	2
I0.M2.m0:region	2
I0.M3.m0:region	2
I0.M4.m0:region	2
I0.M7.m0:region	2
I0.M8.m0:region	2
I0.M9.m0:region	2

Vdsat

VIDAC	0.000
I0.M0.m0:vdsat (V)	104.3E-3
I0.M1.m0:vdsat (V)	104.3E-3
I0.M2.m0:vdsat (V)	-97.19E-3
I0.M3.m0:vdsat (V)	-97.19E-3
I0.M4.m0:vdsat (V)	-118.7E-3
I0.M7.m0:vdsat (V)	-117.6E-3
I0.M8.m0:vdsat (V)	-118.3E-3
I0.M9.m0:vdsat (V)	104.5E-3

vds

VIDAC	0.000
I0.M0.m0:vds (V)	727.6E-3
I0.M1.m0:vds (V)	727.6E-3
I0.M2.m0:vds (V)	-718.9E-3
I0.M3.m0:vds (V)	-718.9E-3
I0.M4.m0:vds (V)	-850.5E-3
I0.M7.m0:vds (V)	-353.4E-3
I0.M8.m0:vds (V)	-692.7E-3
I0.M9.m0:vds (V)	1.107

vgs

VIDAC	0.000
I0.M0.m0:vgs (V)	727.6E-3
I0.M1.m0:vgs (V)	727.6E-3
I0.M2.m0:vgs (V)	-846.6E-3
I0.M3.m0:vgs (V)	-846.6E-3
I0.M4.m0:vgs (V)	-850.5E-3
I0.M7.m0:vgs (V)	-850.5E-3
I0.M8.m0:vgs (V)	-850.5E-3
I0.M9.m0:vgs (V)	727.6E-3



Rout

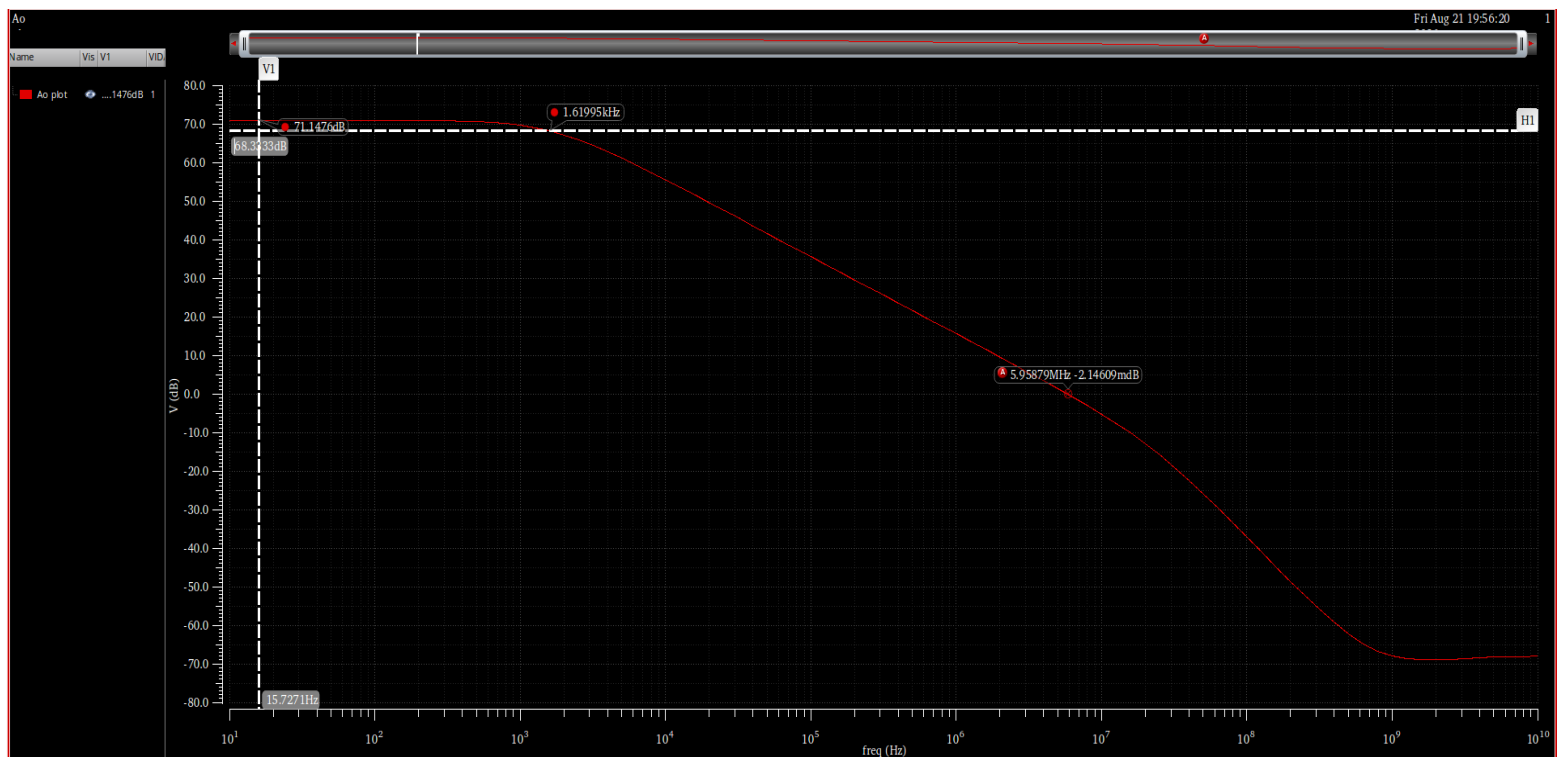
1	
VIDAC	0.000
I0.M0.m0:rout (Ohm)	1.204E6
I0.M1.m0:rout (Ohm)	1.204E6
I0.M2.m0:rout (Ohm)	4.980E6
I0.M3.m0:rout (Ohm)	4.980E6
I0.M4.m0:rout (Ohm)	844.3E3
I0.M7.m0:rout (Ohm)	592.7E3
I0.M8.m0:rout (Ohm)	154.3E3
I0.M9.m0:rout (Ohm)	127.2E3

Diff small signal ccs

Ao, BW and GBW

lab9:TB:1	Ao	3.609k
lab9:TB:1	Ao_dB	71.15
lab9:TB:1	BW	1.698k
lab9:TB:1	fu	5.987M
lab9:TB:1	GBW	6.126M

Plot diff gain (in dB) vs frequency





Hand analysis

$$R_1 = r_{o1} \parallel r_{o3} \\ = (1.204 \text{ M}\Omega) \parallel (4.980 \text{ M}\Omega) = 969.6 \text{ k}\Omega$$

$$A_1 \approx g_{m2} R_1 \\ = (71.82 \mu\text{S})(969.6 \text{ k}\Omega) = 69.64$$

$$R_2 = r_{o8} \parallel r_{o9} \\ = (154.3 \text{ k}\Omega) \parallel (127.2 \text{ k}\Omega) = 69.72 \text{ k}\Omega$$

$$A_2 \approx g_{m9} R_2 \\ = (747.5 \mu\text{S})(69.72 \text{ k}\Omega) = 52.12$$

$$A_0 \approx A_1 A_2 \\ = (69.64)(52.12) = 3.629 \times 10^3 \text{ V/V}$$

$$A_{0,\text{dB}} = 20 \log_{10}(A_0) = 71.20 \text{ dB}$$

$$C_{1,\text{eq}} \approx C_C(1 + A_2) \\ = (1.8 \text{ pF})(1 + 52.12) = 95.61 \text{ pF}$$

$$BW \approx \frac{1}{2\pi R_1 C_{1,\text{eq}}} = 1.717 \text{ kHz}$$

$$GBW \approx A_0(BW) \\ = (3.629 \times 10^3)(1.717 \text{ kHz}) = 6.231 \text{ MHz}$$

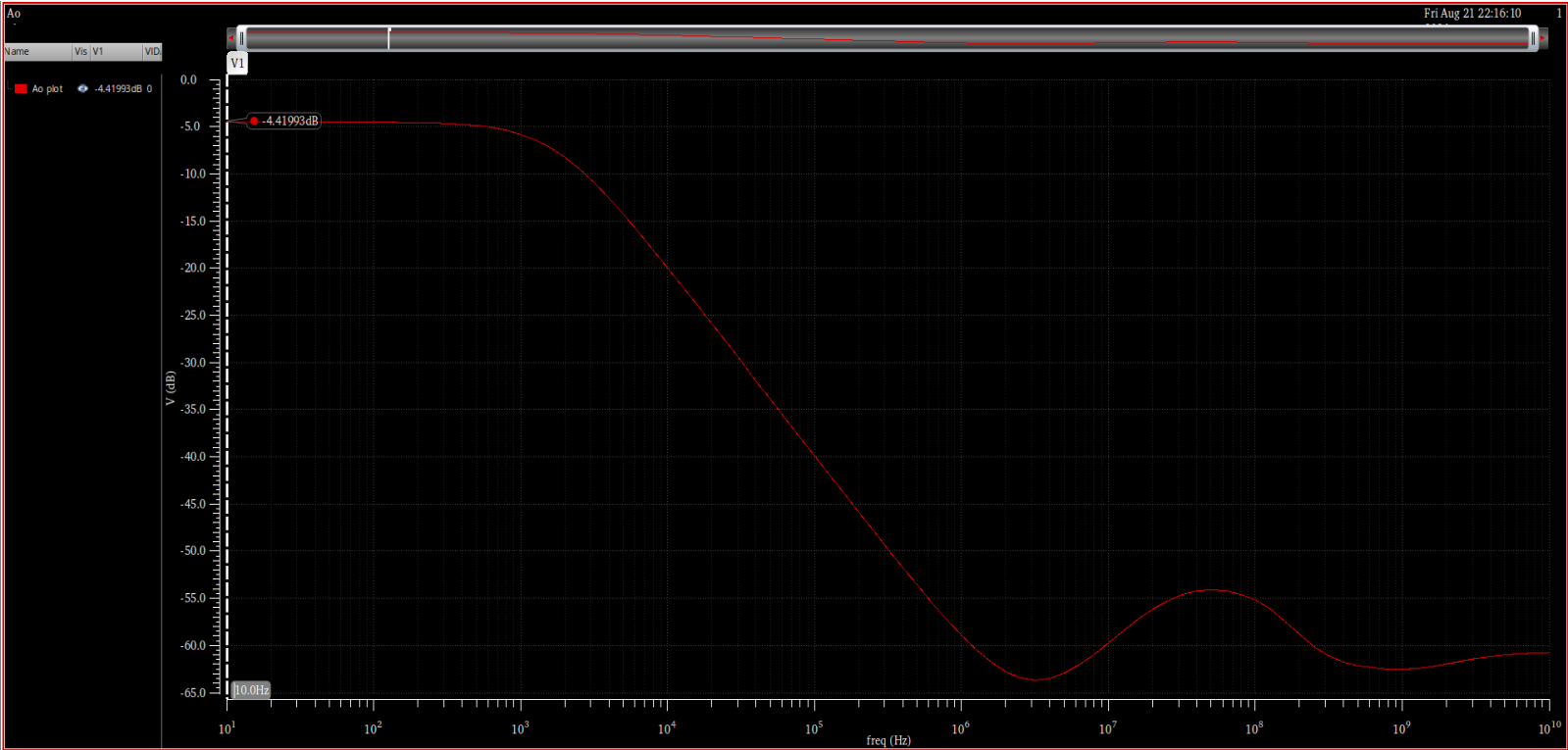
$$UGF \approx \frac{g_{m2}}{2\pi C_C} \\ = \frac{71.82 \mu\text{S}}{2\pi(1.8 \text{ pF})} = 6.350 \text{ MHz}$$

Parameter	Hand analysis	Simulation	Difference
(A_0)	3.629 kV/V	3.609 kV/V	0.56%
(A_0) in dB	71.20 dB	71.15 dB	0.05 dB
Bandwidth	1.717 kHz	1.698 kHz	1.10%
GBW	6.231 MHz	6.126 MHz	1.68%
UGF	6.350 MHz	5.987 MHz	5.72%



CM small signal ccs

AVcm plot



lab9:TB:1	Ao	601.2m
lab9:TB:1	Ao_dB	-4.42
lab9:TB:1	BW	1.698k
lab9:TB:1	GBW	1.021k



Hand analysis

$$R_{SS} = r_{o7} = 592.7 \text{ k}\Omega$$

$$\begin{aligned} A_{CM1} &\approx -\frac{1}{1 + 2g_{m2}R_{SS}} \\ &= -\frac{1}{1 + 2(71.82 \mu\text{S})(592.7 \text{ k}\Omega)} \\ &= -0.01161 \end{aligned}$$

$$\begin{aligned} R_2 &= r_{o8} \parallel r_{o9} \\ &= (154.3 \text{ k}\Omega) \parallel (127.2 \text{ k}\Omega) \\ &= 69.72 \text{ k}\Omega \end{aligned}$$

$$\begin{aligned} A_2 &\approx -g_{m9}R_2 \\ &= -(747.5 \mu\text{S})(69.72 \text{ k}\Omega) \\ &= -52.12 \end{aligned}$$

$$\begin{aligned} A_{CM,0} &\approx A_{CM1}A_2 \\ &= (-0.01161)(-52.12) \\ &= 0.6051 \text{ V/V} \end{aligned}$$

$$\begin{aligned} A_{CM,0,\text{dB}} &= 20 \log_{10}(A_{CM,0}) \\ &= -4.364 \text{ dB} \end{aligned}$$

$$\begin{aligned} C_{1,\text{eq}} &\approx C_C(1 + |A_2|) \\ &= (1.8 \text{ pF})(1 + 52.12) \\ &= 95.61 \text{ pF} \end{aligned}$$

$$\begin{aligned} BW_{CM} &\approx \frac{1}{2\pi R_1 C_{1,\text{eq}}} \\ &= \frac{1}{2\pi(969.6 \text{ k}\Omega)(95.61 \text{ pF})} \\ &= 1.717 \text{ kHz} \end{aligned}$$

$$\begin{aligned} GBW_{CM} &\approx A_{CM,0}BW_{CM} \\ &= (0.6051)(1.717 \text{ kHz}) \\ &= 1.039 \text{ kHz} \end{aligned}$$

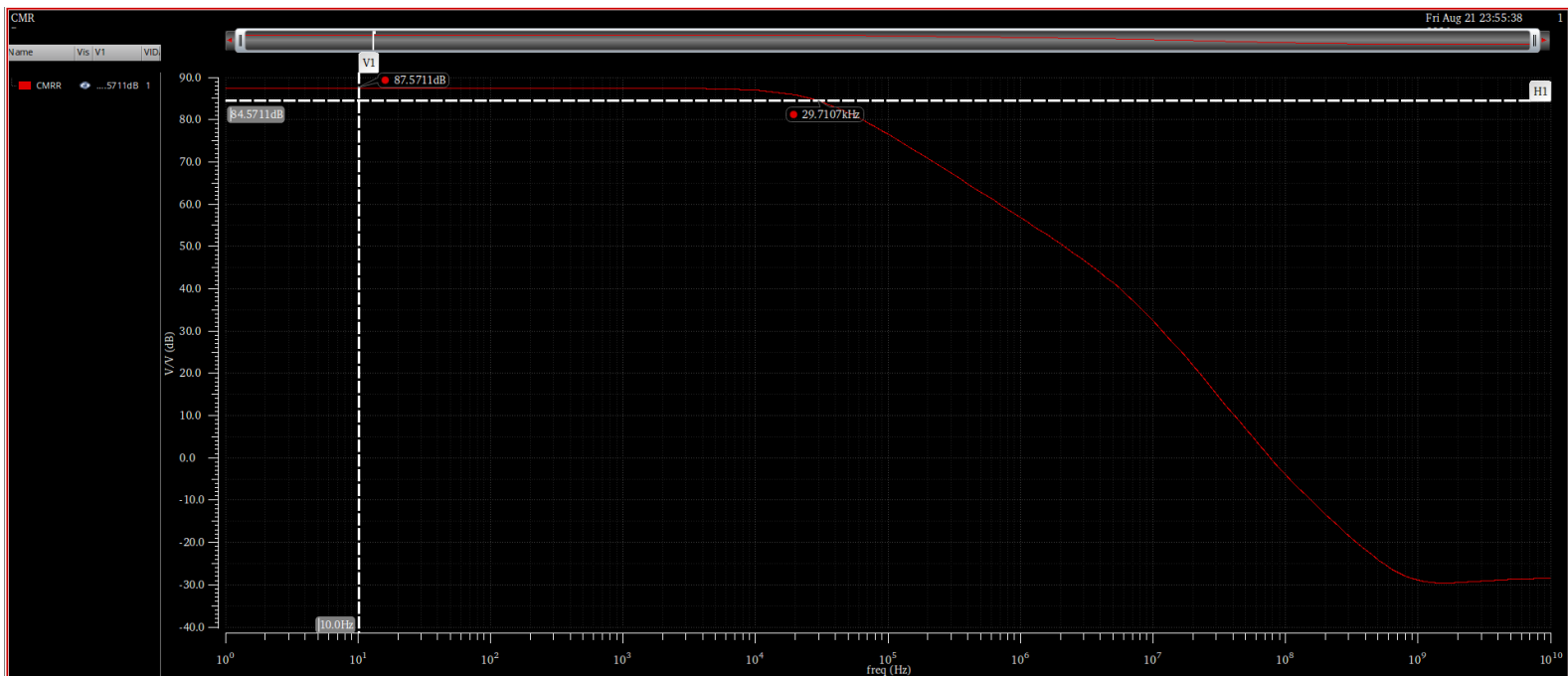


Comparison

Parameter	Hand analysis	Simulation	Difference
(A_CM)	0.6051 V/V	0.6012 V/V	0.64%
(A_CM) in dB	-4.364 dB	-4.42 dB	0.056 dB
CM bandwidth	1.717 kHz	1.698 kHz	1.10%
CM GBW	1.039 kHz	1.021 kHz	1.71%

(Optional) CMRR using XF

CMRR plot



Hand analysis

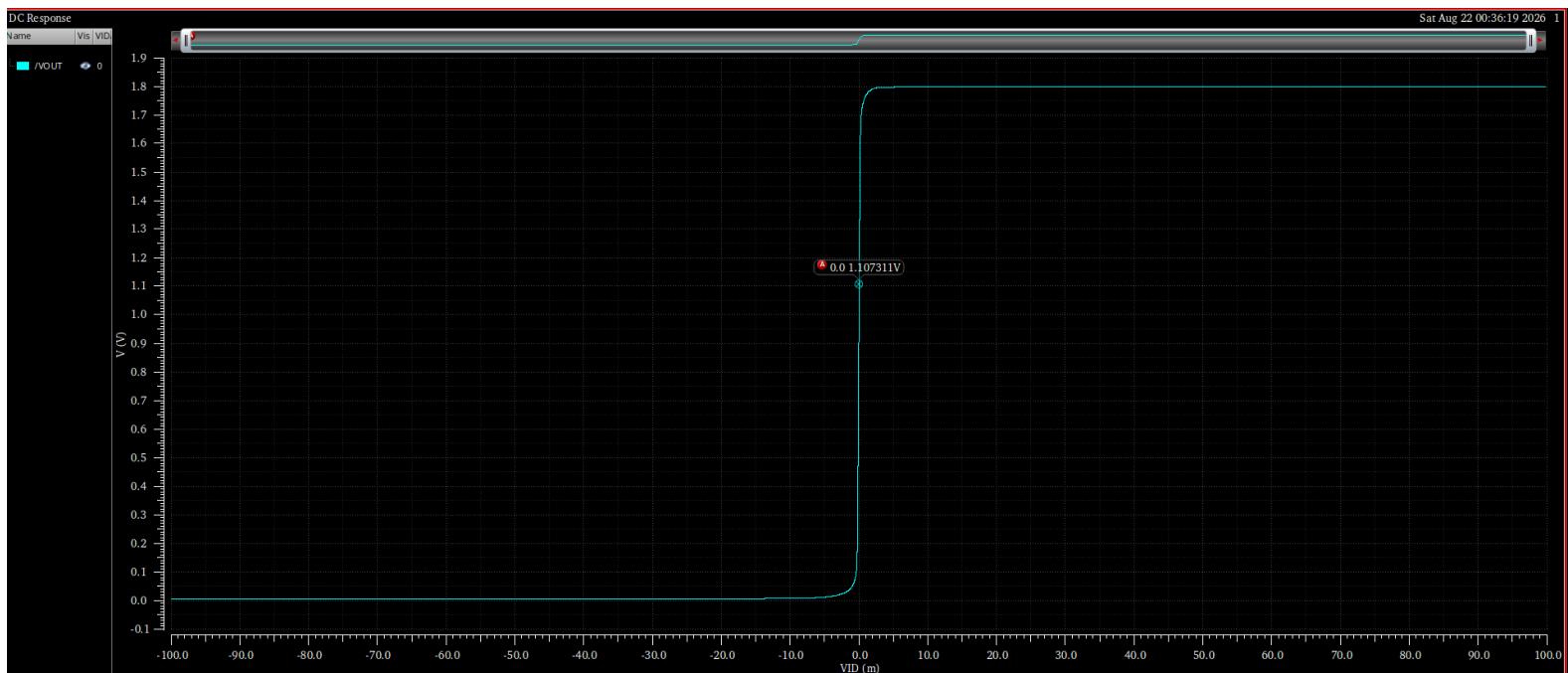
$$\text{CMRR} = 20 \log_{10} \left| \frac{A_{DM}}{A_{CM}} \right| = 20 \log_{10} \left(\frac{3609}{0.6012} \right) = 75.57 \text{ dB}$$

simulation	Hand analysis
87dB	75dB



(Optional) Diff large signal ccs

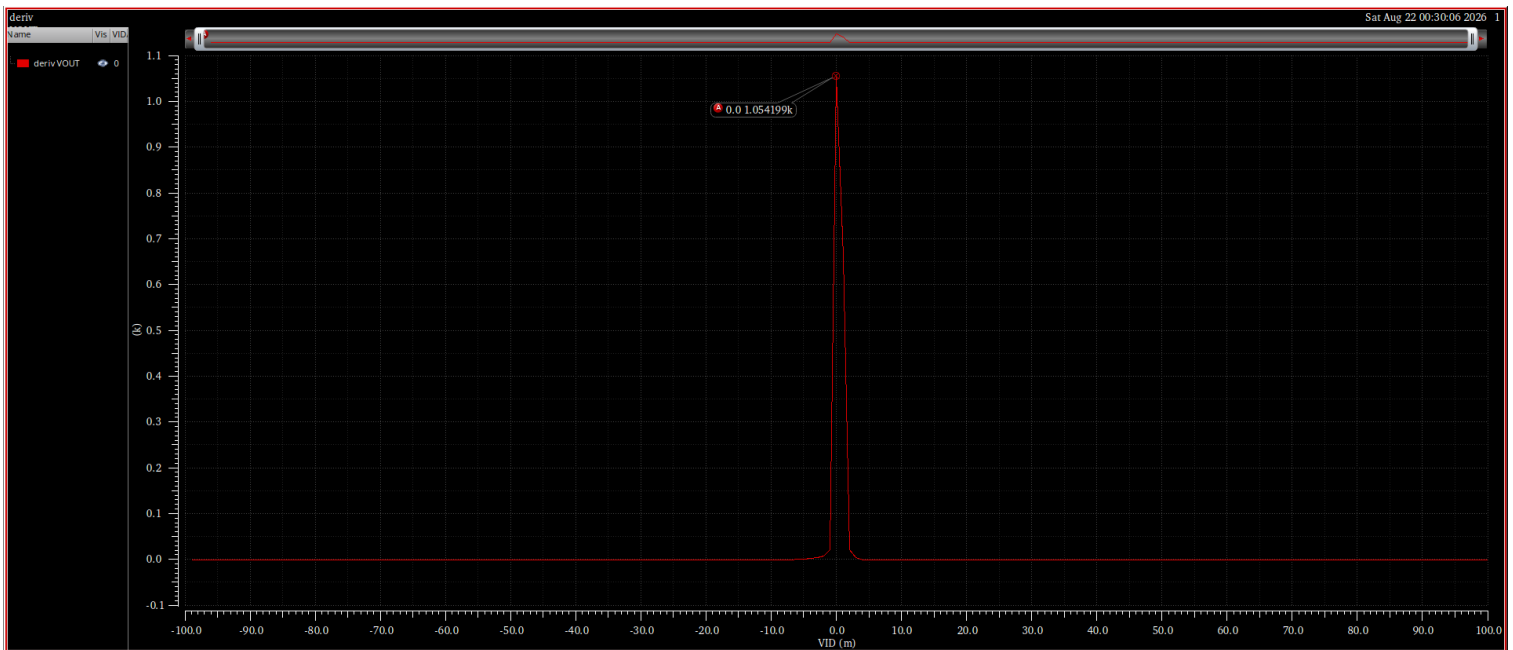
Plot VOUT vs VID.



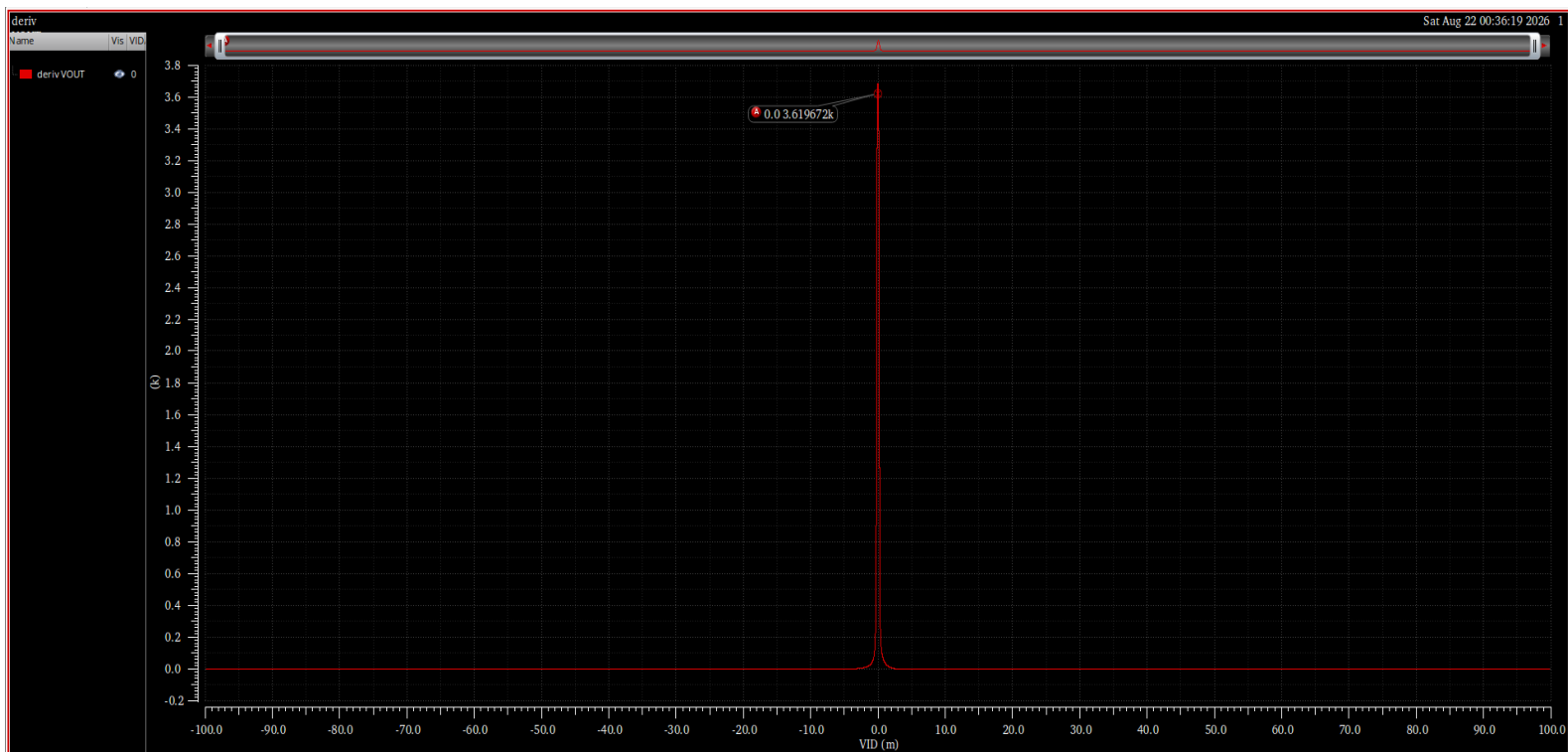
The value from the plot is 1.107 which is identical to the DC-op as the step is very small so this sweep is accurate.



Plot the derivative of VOUT vs VID



The gain is smaller than what we saw in ac analysis but this due to the step size I decreased the step size in another run and this what I got ,the step used is 10u.





CM large signal ccs (region vs VICM)

Plot “region” OP parameter vs VICM



CMIR here is from zero to 840mv

Hand analysis

Lower limit :The VGS of the load is 733mv and the Vth of the differential pair is 787mv so inorder for the input pair to go out of saturation the gate must go below the 0 untill $733-787=-54\text{mv}$ so VDG becomes greater than Vth

Upper limit :the CM device has a gm/id of 14 so $v^*=142\text{mv}$ so to go into sat it need the input to reach $1.8-\text{vgs}_1-\text{vin}<0.142$ and vgs input is roughly 850mv so it needs vin to reach 808mv

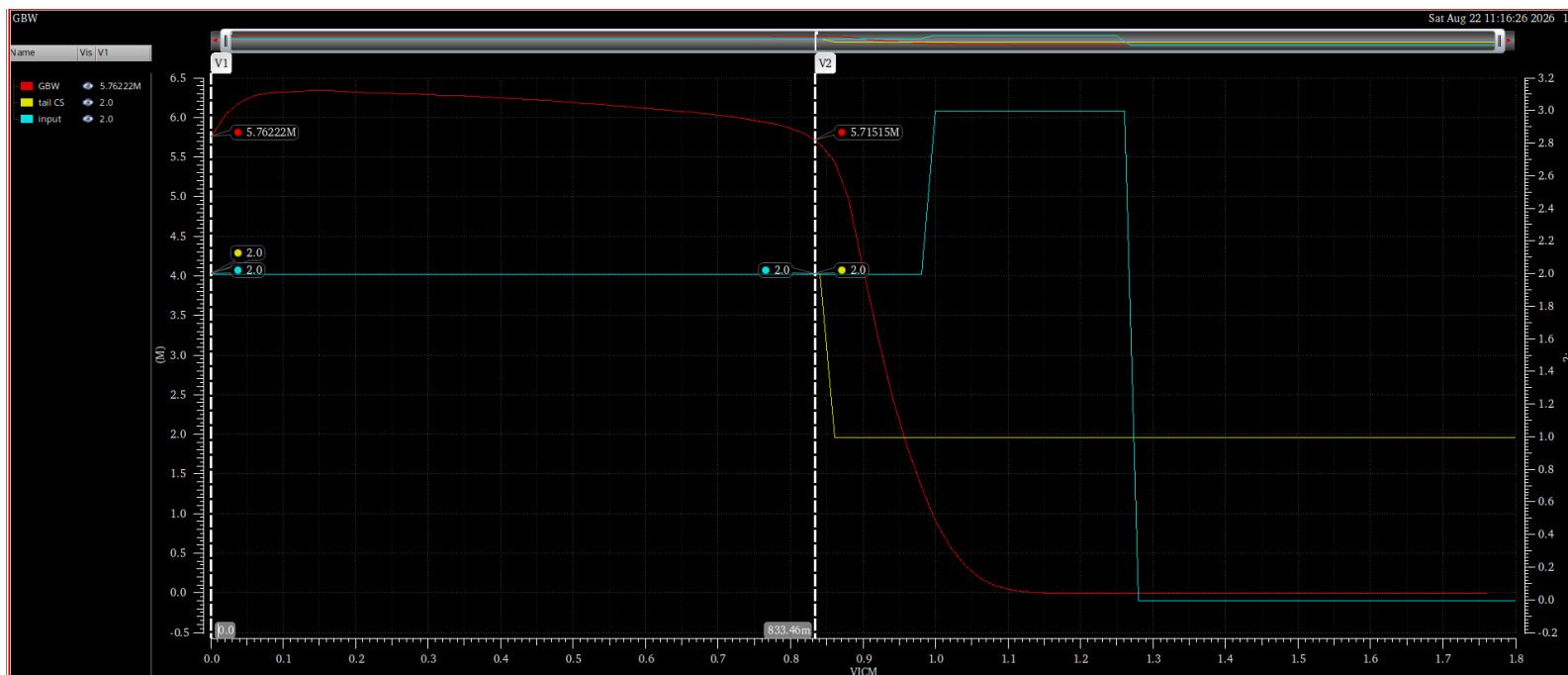
So CMIR is 862mv



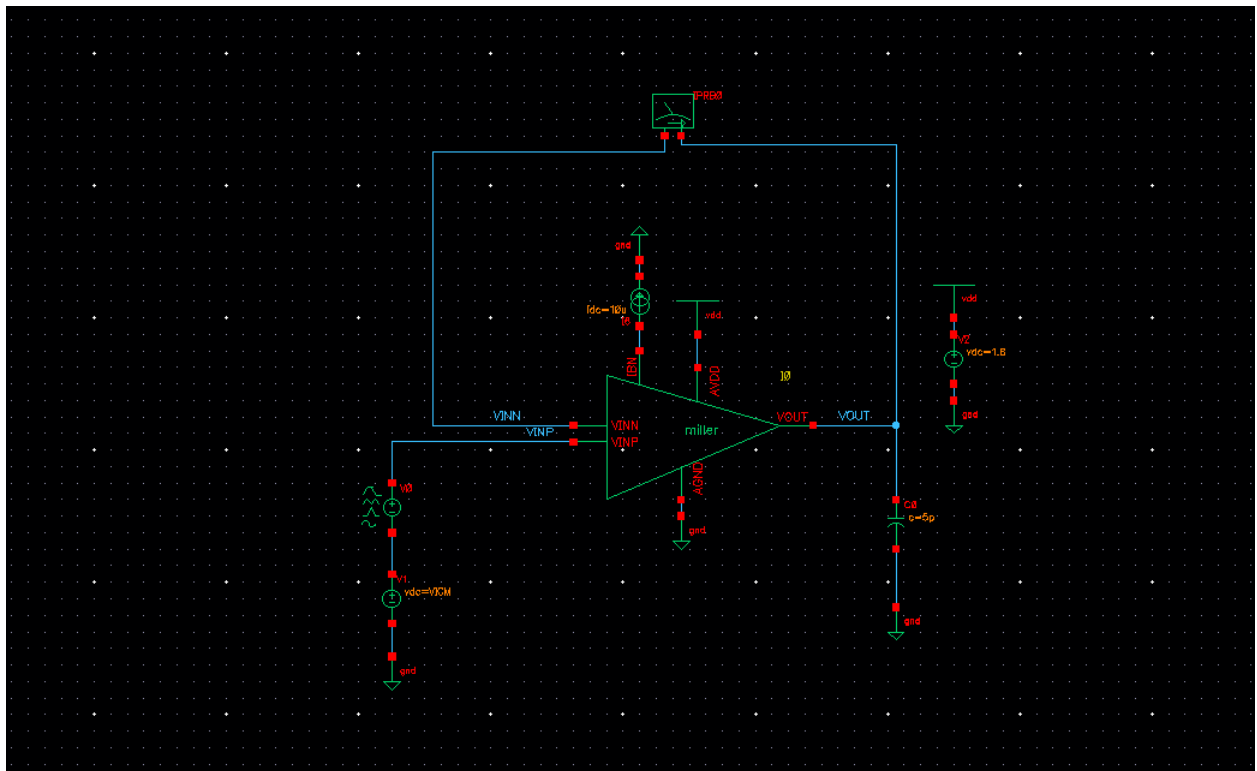
simulation	Hand analysis
840mv	862mv

(Optional) CM large signal ccs (GBW vs VICM)

Plot GBW vs VICM. Plot the results overlaid on the results of the previous method



CMIR =833mv





Gm

1	
VIN	0.000
I0.M0.m0:gm (S)	71.22E-6
I0.M1.m0:gm (S)	71.32E-6
I0.M2.m0:gm (S)	71.78E-6
I0.M3.m0:gm (S)	71.86E-6
I0.M4.m0:gm (S)	141.7E-6
I0.M7.m0:gm (S)	133.2E-6
I0.M8.m0:gm (S)	750.1E-6
I0.M9.m0:gm (S)	778.0E-6

VGS

1	
VIN	0.000
I0.M0.m0:vgs (V)	727.6E-3
I0.M1.m0:vgs (V)	727.6E-3
I0.M2.m0:vgs (V)	-846.5E-3
I0.M3.m0:vgs (V)	-846.6E-3
I0.M4.m0:vgs (V)	-850.5E-3
I0.M7.m0:vgs (V)	-850.5E-3
I0.M8.m0:vgs (V)	-850.5E-3
I0.M9.m0:vgs (V)	737.2E-3

Id

1	
VIN	0.000
I0.M0.m0:id (A)	4.658E-6
I0.M1.m0:id (A)	4.666E-6
I0.M2.m0:id (A)	-4.658E-6
I0.M3.m0:id (A)	-4.666E-6
I0.M4.m0:id (A)	-10.00E-6
I0.M7.m0:id (A)	-9.324E-6
I0.M8.m0:id (A)	-53.27E-6
I0.M9.m0:id (A)	53.27E-6

Region

1	
VIN	0.000
I0.M0.m0:region	2
I0.M1.m0:region	2
I0.M2.m0:region	2
I0.M3.m0:region	2
I0.M4.m0:region	2
I0.M7.m0:region	2
I0.M8.m0:region	2
I0.M9.m0:region	2

Vdsat

1	
VIN	0.000
I0.M0.m0:vdsat (V)	104.2E-3
I0.M1.m0:vdsat (V)	104.3E-3
I0.M2.m0:vdsat (V)	-97.16E-3
I0.M3.m0:vdsat (V)	-97.24E-3
I0.M4.m0:vdsat (V)	-118.7E-3
I0.M7.m0:vdsat (V)	-117.6E-3
I0.M8.m0:vdsat (V)	-119.5E-3
I0.M9.m0:vdsat (V)	108.6E-3

VDS

1	
VIN	0.000
I0.M0.m0:vds (V)	727.6E-3
I0.M1.m0:vds (V)	737.2E-3
I0.M2.m0:vds (V)	-719.1E-3
I0.M3.m0:vds (V)	-709.4E-3
I0.M4.m0:vds (V)	-850.5E-3
I0.M7.m0:vds (V)	-353.4E-3
I0.M8.m0:vds (V)	-1.200
I0.M9.m0:vds (V)	600.1E-3

ROUT

1	
VIN	0.000
I0.M0.m0:rout (Ohm)	1.205E6
I0.M1.m0:rout (Ohm)	1.207E6
I0.M2.m0:rout (Ohm)	4.984E6
I0.M3.m0:rout (Ohm)	4.933E6
I0.M4.m0:rout (Ohm)	844.3E3
I0.M7.m0:rout (Ohm)	592.6E3
I0.M8.m0:rout (Ohm)	180.6E3
I0.M9.m0:rout (Ohm)	105.4E3



Are the DC voltages at the input terminals of the op-amp exactly equal? Why

They are very close due to large open loop gain but not exactly.

Is the DC voltage at the output of the first stage exactly equal to the value in the open-loop simulation? Why

It close but there is a difference around 10mv due to that vout biases the circuit that the vout is exactly what is needed to keep the output very close to the input so it creates like a slight steer in the differential amp as the inputs are slightly different thus the currents and the node voltages

Is the current (and gm) in the input pair exactly equal? Why?

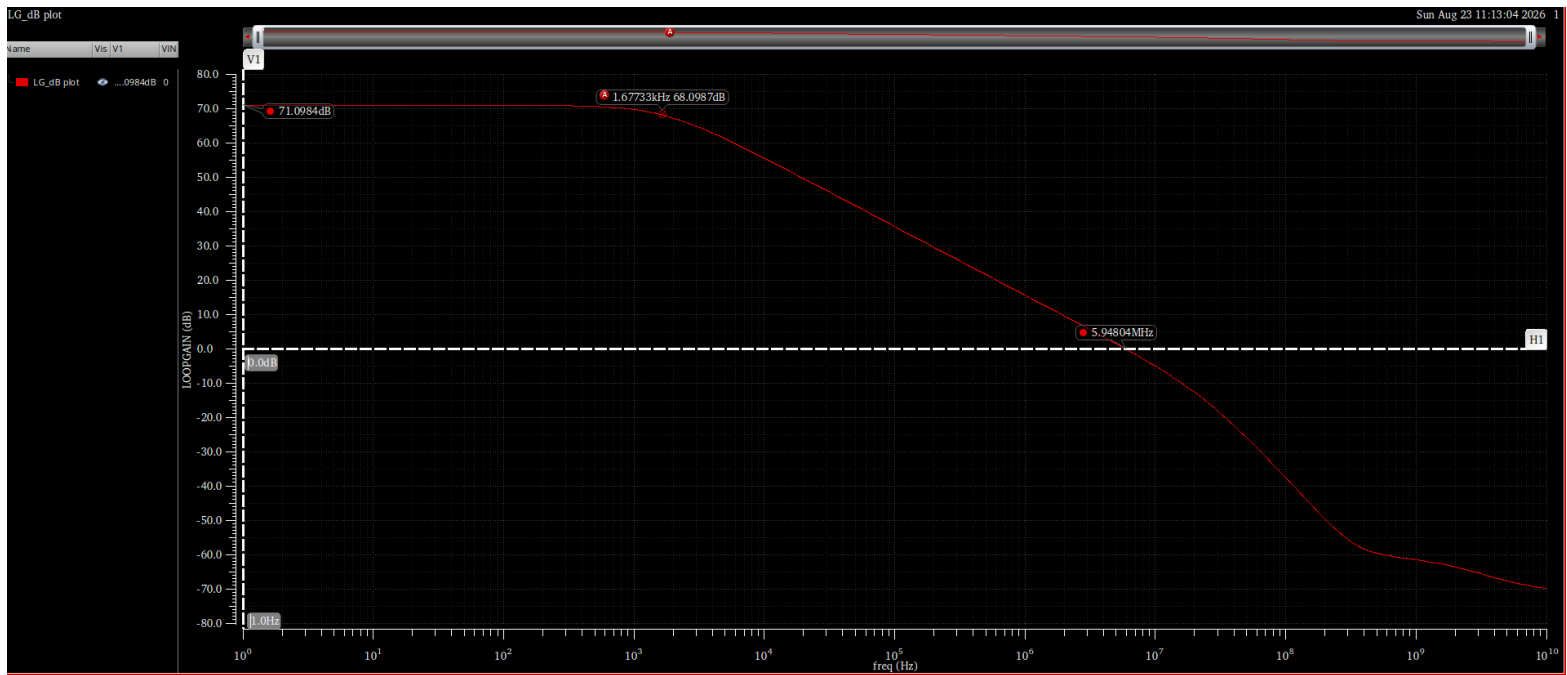
No as we said earlier the input at each side is slightly different so there can be a slight steer due the biasing of vout to keep as the input so the currents differs and so the gms.



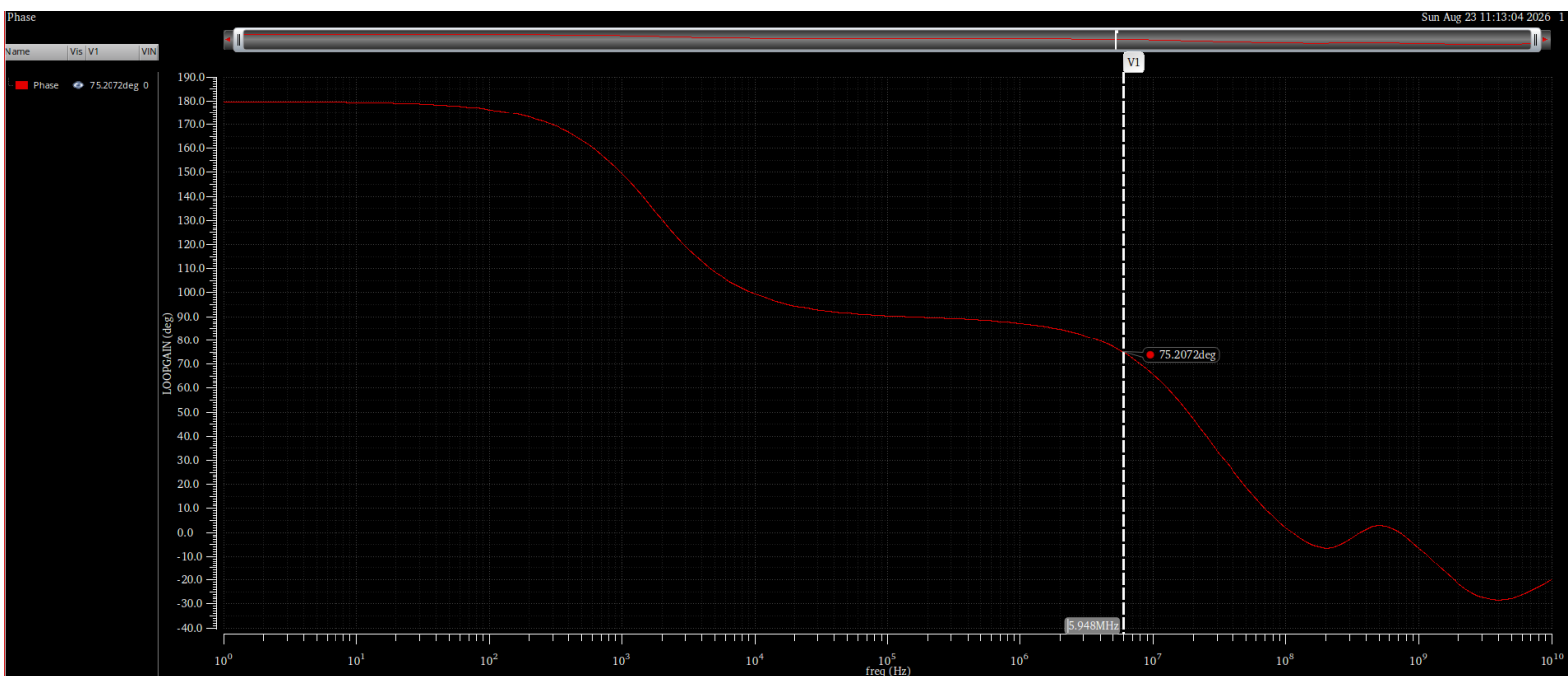
Loop gain

Plot loop gain in dB and phase vs frequency

LG



Phase





lab9:TBCL:1	LG	3.589k
lab9:TBCL:1	LG_dB	71.1
lab9:TBCL:1	BW	1.703k
lab9:TBCL:1	UGF	5.958M
lab9:TBCL:1	GBW	6.112M

Compare DC gain, fu, and GBW with those obtained from open-loop simulation. Comment

CL

lab9:TBCL:1	LG	3.589k
lab9:TBCL:1	LG_dB	71.1
lab9:TBCL:1	BW	1.703k
lab9:TBCL:1	UGF	5.958M
lab9:TBCL:1	GBW	6.112M

OL

lab9:TB:1	Ao	3.609k
lab9:TB:1	Ao_dB	71.15
lab9:TB:1	BW	1.698k
lab9:TB:1	fu	5.987M
lab9:TB:1	GBW	6.126M

As we expected the results are close and beta =1 so LG =AOL but

The gain slightly decreased and this due to the feedback trying to change the biasing a bit so vout = vin

Report PM

VIN	/phaseMargin (Deg)
1 0.000	75.23

Hand calculation

$PM = 90 - \arctan(WU/WP2)$, $wu = gm1/cc$ $wp2 = gm2/cL$

$Wu = 71.86\mu S / 2pF = 35.93Mrad/s$

$Wp2 = 778\mu S / 5pf = 155.6Mrad/s$

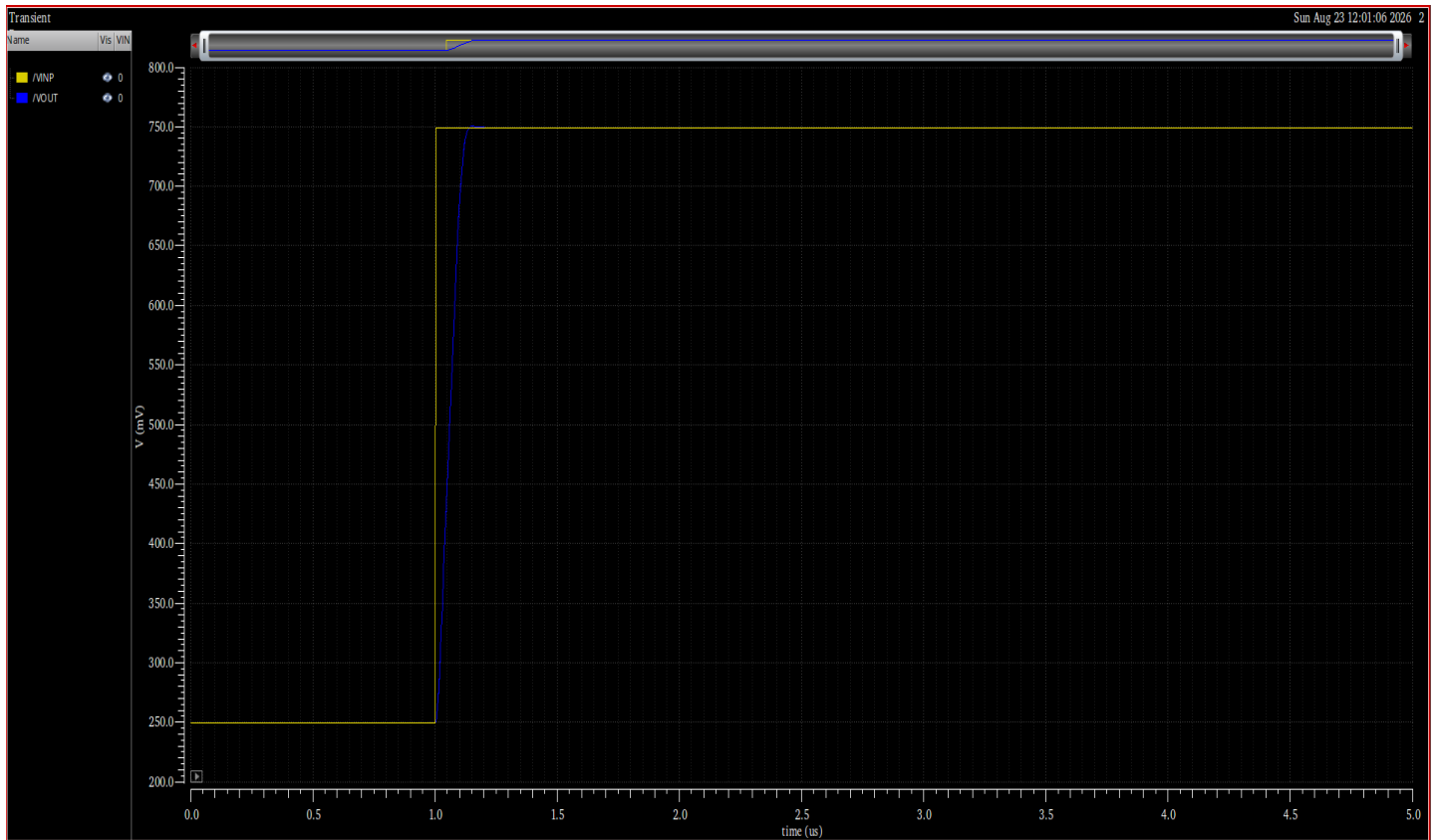
$90 - \arctan (35.93/155.6) = 77deg$

Simulation	Hand analysis
75.23	77

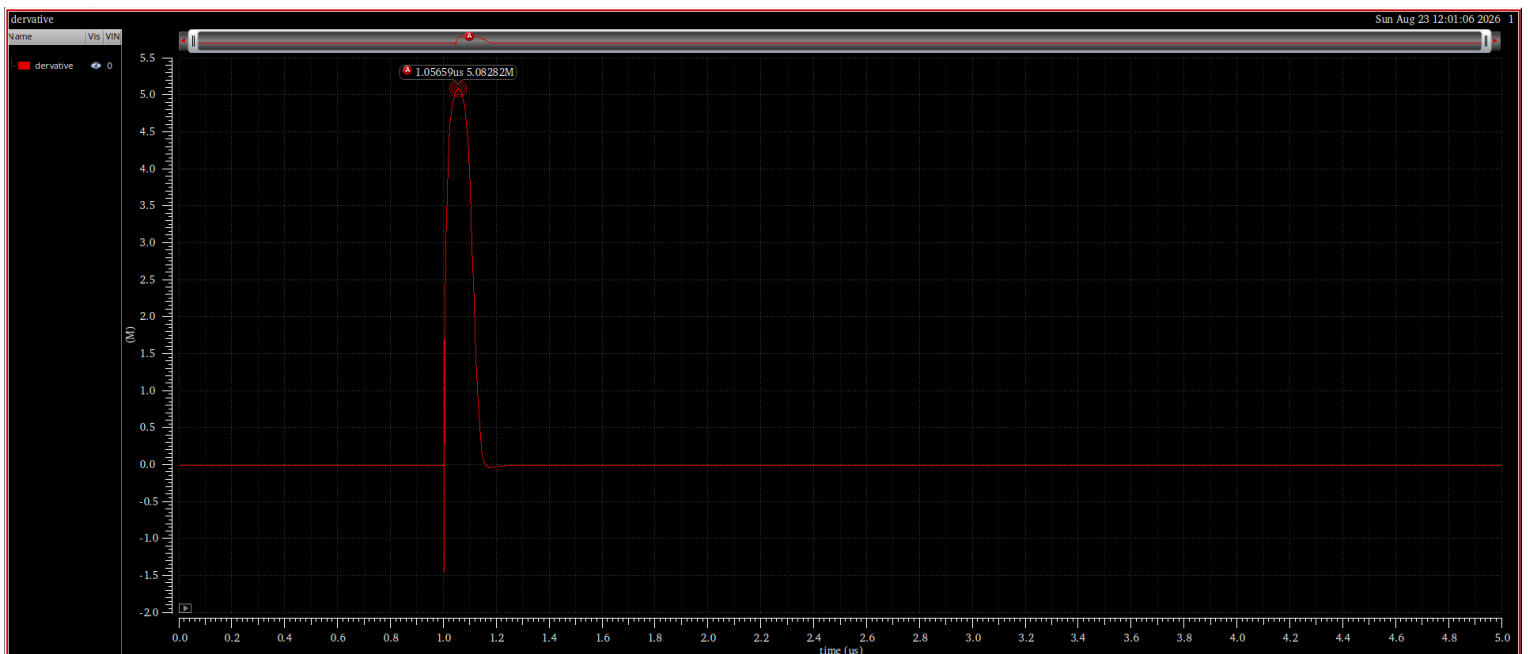


Slew rate

Report Vin and Vout overlaid



Slew rate using derivative for vout





Hand analysis

$$\text{Slew rate} = I_{B1}/C_C = 10\mu/2\text{pf} = 5\text{V}/\mu\text{S}$$

There is an important note I had to change the C_C from 1.8 to 1.7 to achieve this slew rate and it did not affect any other spec it only increases the slew rate from 4.9 to 5 and the phase margin decreased one degree but still above the spec

simulation	Hand analysis
5.08V/ μ s	5V/ μ s

Settling time

lab9:TBCL:1	settling time error 10%	46.43n
lab9:TBCL:1	settling time error 5%	54.1n
lab9:TBCL:1	Rise Time 10 %to 90%	38.8n

Hand analysis

$$\text{Rise time} = 2.2 \tau = 2.2 * C_C / g_{m1} = 2.2 * 2\text{pf} / 71.87\mu\text{S} = 61.2\text{ns}$$

simulation	Hand analysis
38.8ns	61.2ns

Why is the difference so large because we approximated it in hand calculations as a single pole system but is a second order system and second order systems are always faster than single ones

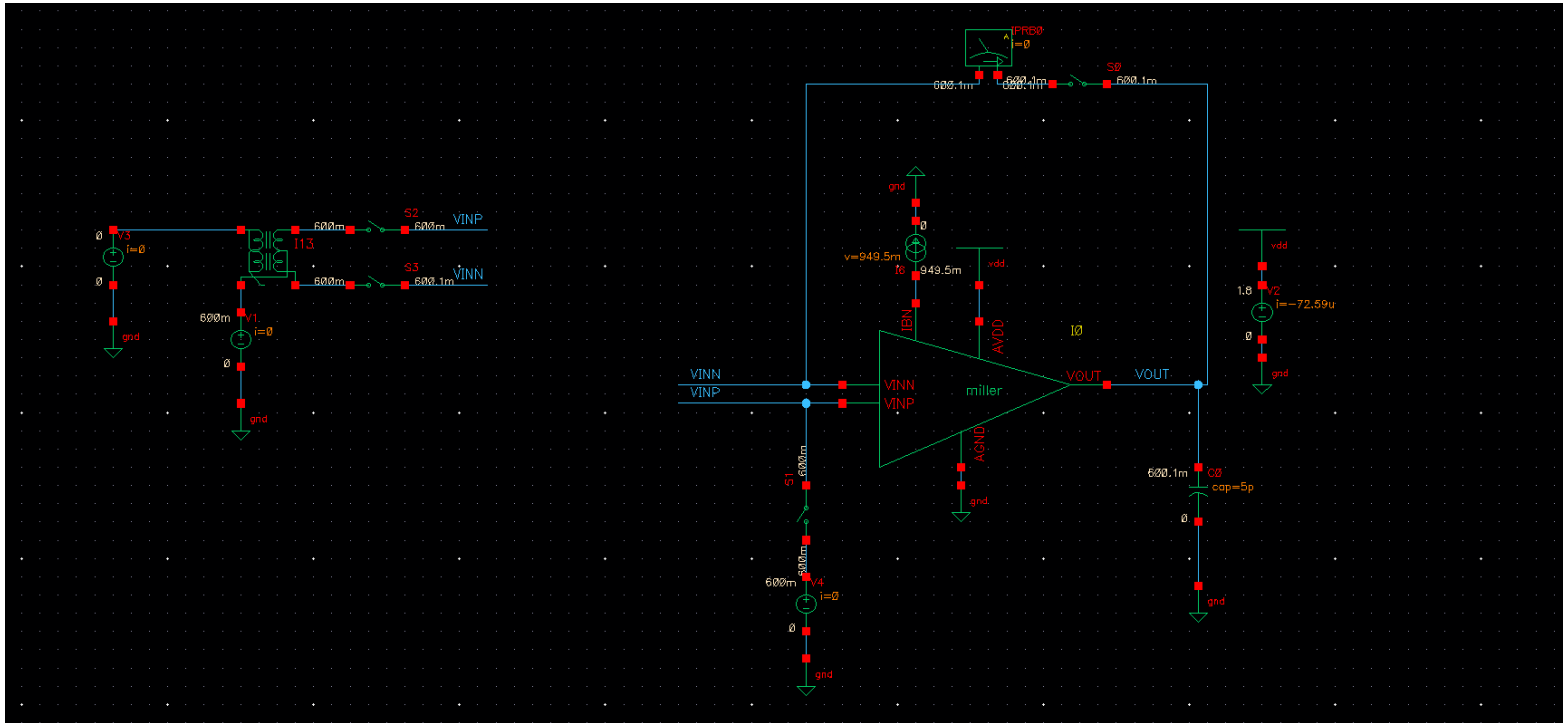
Do you see any ringing? Why?

There is ringing because its not ideally critically damped it is still slightly underdamped system even the phase margin ensure this as it falls below 76 by a small margin

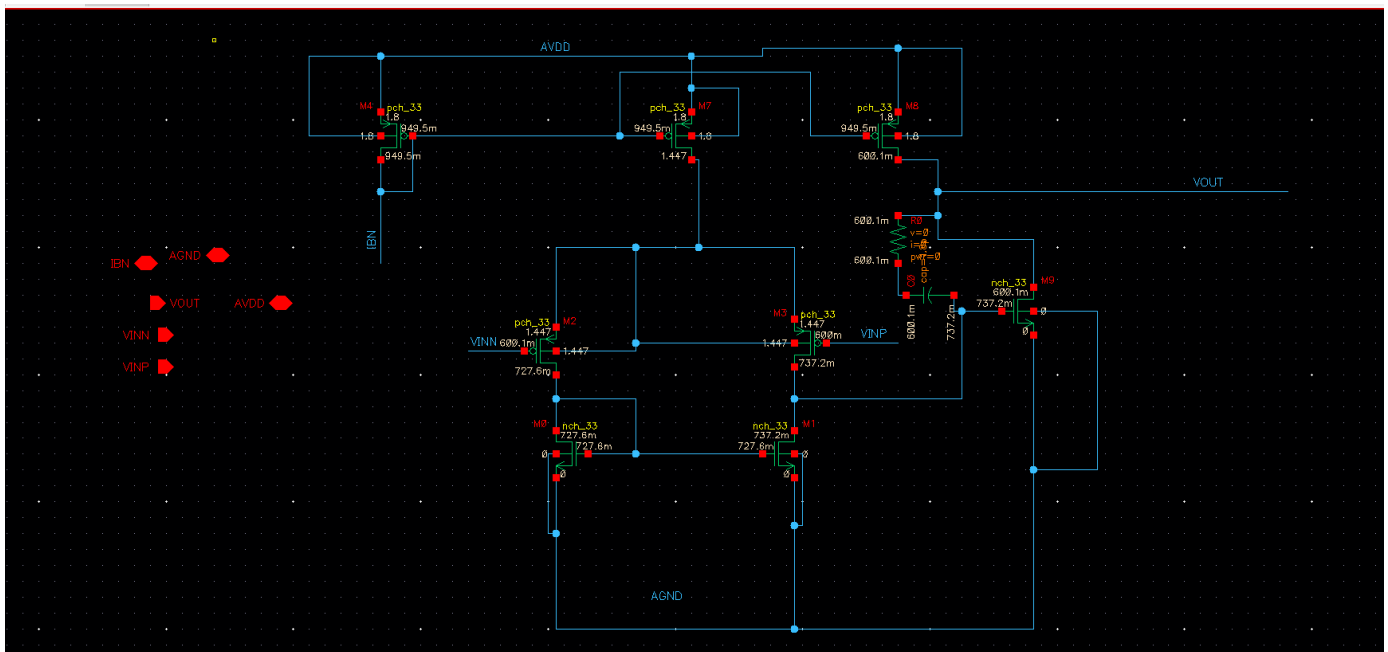


Part 5 (optional): DC Closed Loop AC Open-Loop OTA Simulation

Schematic



DCOP





I0.M0.m0:gm I0.M1.m0:gm I0.M2.m0:..	
1	
VIN	0.000
I0.M0.m0:gm (S)	71.22E-6
I0.M1.m0:gm (S)	71.32E-6
I0.M2.m0:gm (S)	71.78E-6
I0.M3.m0:gm (S)	71.86E-6
I0.M4.m0:gm (S)	141.7E-6
I0.M7.m0:gm (S)	133.2E-6
I0.M8.m0:gm (S)	750.1E-6
I0.M9.m0:gm (S)	778.0E-6

1	
VIN	0.000
I0.M0.m0:id (A)	4.658E-6
I0.M1.m0:id (A)	4.666E-6
I0.M2.m0:id (A)	-4.658E-6
I0.M3.m0:id (A)	-4.666E-6
I0.M4.m0:id (A)	-10.00E-6
I0.M7.m0:id (A)	-9.324E-6
I0.M8.m0:id (A)	-53.27E-6
I0.M9.m0:id (A)	53.27E-6

1	
VIN	0.000
I0.M0.m0:vgs (V)	727.6E-3
I0.M1.m0:vgs (V)	727.6E-3
I0.M2.m0:vgs (V)	-846.5E-3
I0.M3.m0:vgs (V)	-846.6E-3
I0.M4.m0:vgs (V)	-850.5E-3
I0.M7.m0:vgs (V)	-850.5E-3
I0.M8.m0:vgs (V)	-850.5E-3
I0.M9.m0:vgs (V)	737.2E-3

I0.M0.m0:vds I0.M1.m0:vds I0.M2.	
1	
VIN	0.000
I0.M0.m0:vds (V)	727.6E-3
I0.M1.m0:vds (V)	737.2E-3
I0.M2.m0:vds (V)	-719.1E-3
I0.M3.m0:vds (V)	-709.4E-3
I0.M4.m0:vds (V)	-850.5E-3
I0.M7.m0:vds (V)	-353.4E-3
I0.M8.m0:vds (V)	-1.200
I0.M9.m0:vds (V)	600.1E-3

1	
VIN	0.000
I0.M0.m0:vdsat (V)	104.2E-3
I0.M1.m0:vdsat (V)	104.3E-3
I0.M2.m0:vdsat (V)	-97.16E-3
I0.M3.m0:vdsat (V)	-97.24E-3
I0.M4.m0:vdsat (V)	-118.7E-3
I0.M7.m0:vdsat (V)	-117.6E-3
I0.M8.m0:vdsat (V)	-119.5E-3
I0.M9.m0:vdsat (V)	108.6E-3

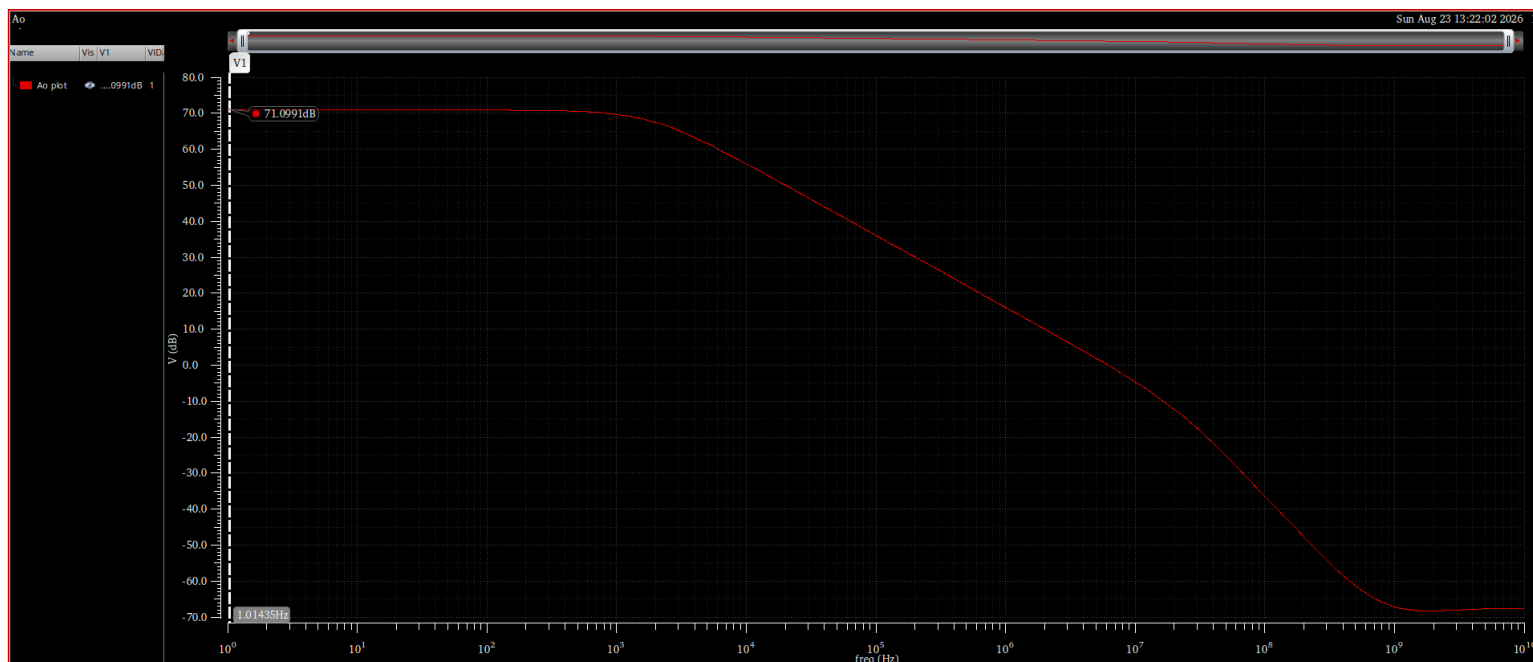
1	
VIN	0.000
I0.M0.m0:region	2
I0.M1.m0:region	2
I0.M2.m0:region	2
I0.M3.m0:region	2
I0.M4.m0:region	2
I0.M7.m0:region	2
I0.M8.m0:region	2
I0.M9.m0:region	2

1	
VIN	0.000
I0.M0.m0:rout (Ohm)	1.205E6
I0.M1.m0:rout (Ohm)	1.207E6
I0.M2.m0:rout (Ohm)	4.984E6
I0.M3.m0:rout (Ohm)	4.933E6
I0.M4.m0:rout (Ohm)	844.3E3
I0.M7.m0:rout (Ohm)	592.6E3
I0.M8.m0:rout (Ohm)	180.6E3
I0.M9.m0:rout (Ohm)	105.4E3



AC analysis

AVd plot against frequency



lab9:TB bonus:1	Ao	3.589k
lab9:TB bonus:1	Ao_dB	71.1
lab9:TB bonus:1	BW	1.802k
lab9:TB bonus:1	fu	6.272M
lab9:TB bonus:1	GBW	6.466M